

collapse. But even if carbon were more abundant than oxygen and/or no separation mechanism were operating solidification would still ensure at least partial collapse. This is because when thermonuclear ignition happens in the middle of a solid, the burning propagates by conduction (the successively melted layers only afterwards become convective). The characteristic speeds are between 10^{-4} and 10^{-5} times the local speed of sound¹⁸. Taking into account the new corrections to the electron screening factors¹⁹, the ignition density of the reaction $^{12}\text{C} + ^{12}\text{C}$ becomes $\rho_c \sim 3.5 \times 10^9 \text{ g cm}^{-3}$. For instance, assuming the interior to be solid from the centre out to a radius $5 \times 10^7 \text{ cm}$ (about half the stellar radius), it takes $\sim 500 \text{ s}$ for the conductive burning front to reach the edge of the solid core (in contrast with $\sim 1 \text{ s}$ if we had assumed a convectively-driven front, or $\sim 0.1 \text{ s}$ for a detonation wave). Even at those relatively low ignition densities, the electron captures on the matter in nuclear statistical equilibrium (NSE)²⁰ are fast enough to remove pressure efficiently and start the collapse: $\tau_e = Y_e / \dot{Y}_e \leq 5 \text{ s}$ (where Y_e is the electron mole number). When the burning front reaches the fluid layers, it changes into a convectively-driven front and accelerates.

Calculations were performed assuming central ignition in a white dwarf with a central density of $3 \times 10^9 \text{ g cm}^{-3}$, with a solid core of one-tenth of the total mass, composed of a mixture of equal parts of carbon and oxygen. While the burning propagates through the solid core at a velocity of 2 km s^{-1} hydrostatic equilibrium was assumed. When the burning front reaches the solid/liquid transition, the material between that point and the layer corresponding to $1 M_\odot$ was instantaneously incinerated. Then the hydrodynamical behaviour was followed with an explicit numerical code²¹.

Our results are preliminary because we have extrapolated the approximations used for the matter in NSE to higher densities and lower Y_e than prescribed, and the hydrostatic/hydrodynamic transition should be treated more carefully. In the same way, burning propagation in the solid, as well as in the fluid, layers needs a more detailed treatment. Keeping these restrictions in mind, we conclude that the liquid layers are ejected and the solid/fluid transition sets a lower limit to the mass of the bounded remnant. Due to the low densities of those layers ($\rho \leq 2.35 \times 10^9 \text{ g cm}^{-3}$) the neutron excess of the ejected matter is not too high ($\langle \eta \rangle \sim 3 \times 10^{-3}$).

Thus, the mere solidification of the star's interior appears to determine the implosion of the frozen layers on carbon ignition. The burning front, when reaching the still fluid layers mimics an off-centre ignition. A single parameter (the size of the solid core, resulting from the interplay between the cooling time before mass transfer and the heating effects of the accretion process²²) determines the amount of matter ejected. So, the whole range from mildly explosive collapse (leading to the formation of Type II X-ray sources) to 'fast' and 'slow' SN I might be covered by a single mechanism based on a well understood physical process.

This work was supported in part by the CAICYT of the Ministerio de Education y Ciencia Español and by the CIRIT of the Generalitat de Catalunya.

Received 8 December 1983; accepted 30 March 1984.

- Branch, D. *et al.* *Astrophys. J. Lett.* **252**, L61–L64 (1982).
- Arnett, W. D. *Astrophys. J.* **240**, 105–108 (1980).
- Colgate, S. A., Petschek, A. G. & Kriese, J. T. *Astrophys. J. Lett.* **237**, L81–L85 (1980).
- Axelrod, T. S. thesis, Univ. California, Santa Cruz (1980).
- Wheeler, J. C. *Nature* **302**, 209 (1983).
- Branch, D. *Astrophys. J.* **248**, 1076–1080 (1981).
- Arnett, W. D. in *Supernovae: A Survey of Current Research* (eds M. J. Rees, & Stoneham, R. J.) 221–251 (Reidel, Dordrecht, 1982).
- Wheeler, J. C. in *Supernovae: A Survey of Current Research* (eds Rees M. J. & Stoneham, R. J.) 167–203 (Reidel, Dordrecht, 1982).
- Riegler, G. R. *et al.* *Astrophys. J. Lett.* **248**, L13–L16 (1981).
- Leventhal, M. & MacCallum, C. J. *Proc. Workshop on the Galactic Center* (eds Riegler, G. R. & Blandford, R. D.) 132–138 (California Institute of Technology, 1982).
- Colgate, S. A. & Petschek, A. G. *Nature* **296**, 804–805 (1982).
- Vader, J. P., Van den Heuvel, E. P. J., Lewin, W. H. G. & Takens, R. J. *Astr. Astrophys.* **113**, 328–335 (1982).
- Van den Heuvel, E. P. J. *Space Sci. Rev.* **30**, 623–642 (1981).
- Canal, R., Isern, J. & Labay, J. *Nature* **296**, 225–226 (1982).
- Isern, J., Labay, J., Hernanz, M. & Canal, R. *Astrophys. J.* **273**, 320–329 (1983).

- Schatzman, E. *Cataclysmic Variables and Related Objects*, IAU Colloq., No. 72, 149–153 (1983).
- Kettner, K. U. *et al.* *Z. Phys. A* **308**, 73–84 (1982).
- Buchler, J. R., Colgate, S. A. & Mazurek, T. J. *J. Phys. Suppl.* **41**, C2–159 (1980).
- Ichimaru, S. & Utsumi, K. *Astrophys. J. Lett.* **269**, L51–L55 (1983).
- Epstein, R. E. & Arnett, W. D. *Astrophys. J.* **201**, 202–211 (1975).
- Colgate, S. A. & White, R. H. *Astrophys. J.* **143**, 626–681 (1966).
- Nomoto, K. *Astrophys. J.* **253**, 798–810 (1982).

Topography of the shield volcano, Olympus Mons on Mars

S. S. C. Wu, P. A. Garcia, R. Jordan, F. J. Schafer & B. A. Skiff

United States Geological Survey, 2255 North Gemini Drive, Flagstaff, Arizona 86001, USA

Olympus Mons, one of the largest known shield volcanoes in the Solar System, covers an area of $>3.2 \times 10^5 \text{ km}^2$ and has a diameter of $>600 \text{ km}$, excluding its vast aureole deposits. The structure is five times larger than the largest shield volcano on the Earth. It is situated on the north-west flank of the Tharsis volcanic region, a broad topographic rise on the martian surface. The volcano has three physical subdivisions: the summit caldera, the terraced upper flanks, and the lower flanks, which terminate in a scarp 2–10 km high that nearly surrounds the structure. A large block of images of the Tharsis region, including Olympus Mons, was obtained by the Viking mission¹. Here we present a topographic map of Olympus Mons, compiled using various combinations of stereo pairs of these images, together with stereoscopic perspective views generated by image processing techniques.

The contour map of Olympus Mons was compiled at a scale of 1:1,000,000 with a contour interval of 200 m, using 91 stereoscopic models of Viking Orbiter photographs². Special photogrammetric techniques were used for this compilation. These techniques had been developed to use the Viking Orbiter photographs, which have extremely narrow field-of-view³, for the systematic mapping of Mars. Three hundred and sixteen control points were produced from a block of 103 photographs using analytical aerotriangulation methods. Digital elevation data were collected from stereo models on the analytical plotter at the time of map compilation. Using techniques of digital image processing⁴ shown in Fig. 2, perspective stereoscopic views were made at various vertical viewing angles from different desired azimuths. The summit caldera and the terraced upper flank are viewed monoscopically from the east and the west in Fig. 2a and b, respectively; the scarps of the lower flanks are viewed stereoscopically from the north and the south-east in Fig. 2c and d.

Previous estimates of the height of Olympus Mons ranged from 17 to 23 km above the surrounding area^{5–12}. A height of 27 km above the Mars topographic datum¹³ is shown on the US Geological Survey 1:25 million scale, global topographic map of Mars¹⁴, which was compiled by synthesizing remotely-sensed data from the Mariner 9 Mars mission and Earth-based radar observations^{15–16}. The newly compiled topographic map (Fig. 1), shows the elevation of Olympus Mons to be 26,400 m above the Mars datum. The volume determination of Olympus Mons calculated for individual segments bounded by contour lines on Fig. 1 is given in Table 1.

Olympus Mons has the general form and slope of a classic terrestrial basaltic shield volcano^{8,17,18}. Major topographic features include a complex caldera at the summit that is $\sim 80 \text{ km}$ across, upper flanks that form a concentric pattern of terraced terrain, and a steep basal scarp. The summit caldera is composed of at least five coalesced collapse craters¹⁹. Concentric terraces on the upper flanks are prominent on the east, south-east, west, and south-west sides²⁰. Typically, the terraces are spaced 15–20 km apart. The unique, nearly continuous, basal scarp rises more than 10 km above the basal plains on the north and north-west, (Fig. 1); on the south-east, the scarp rises 4–5 km

Table 1 Volume determination of Olympus Mons

	Elevation (km)	Volume of segment	Accumulated ($\times 10^6 \text{ km}^3$)	Elevation (km)	Volume of segment	Accumulated ($\times 10^6 \text{ km}^3$)	Elevation (km)	Volume of segment	Accumulated ($\times 10^6 \text{ km}^3$)
Above	26	0.0004	0.0004	17	0.073	0.336	9	0.223	1.553
	25	0.006	0.007	16	0.088	0.424	8	0.236	1.789
	24	0.013	0.020	15	0.103	0.527	7	0.247	2.036
	23	0.022	0.042	14	0.120	0.647	6	0.262	2.298
	22	0.029	0.071	13	0.140	0.787	5	0.296	2.594
	21	0.035	0.106	12	0.158	0.945	4	0.365	2.959
	20	0.043	0.149	11	0.180	1.125	3	0.436	3.395
	19	0.052	0.201	10	0.205	1.330	2	0.467	3.862
	18	0.061	0.263						

Volumes were calculated, using a planimeter, for individual segments bounded by contour lines on the map (Fig. 1). The third column lists cumulative volumes, measured downward from the top in increments of 1 km. Blasius^{7,8} reported a total volume of $2.7 \times 10^6 \text{ km}^3$ for Olympus Mons and Beatty²⁵ reported a volume of $3\text{--}4 \times 10^6 \text{ km}^3$. From the new map (Fig. 1) we calculate a total volume of $2.594 \times 10^6 \text{ km}^3$ for the part of the volcano above the 5-km elevation. Assuming a density of 3.0 g cm^{-3} (ref. 26), we estimate the topographic mass of Olympus Mons to be $7.782 \times 10^{18} \text{ kg}$ above the 5-km elevation. If estimated volumes between elevations of 2 and 5 km are added to this calculation, the total volume above the 2-km contour is $3.862 \times 10^6 \text{ km}^3$. This, in turn, gives a total mass of $11.586 \times 10^{18} \text{ kg}$ above the 2-km elevation.

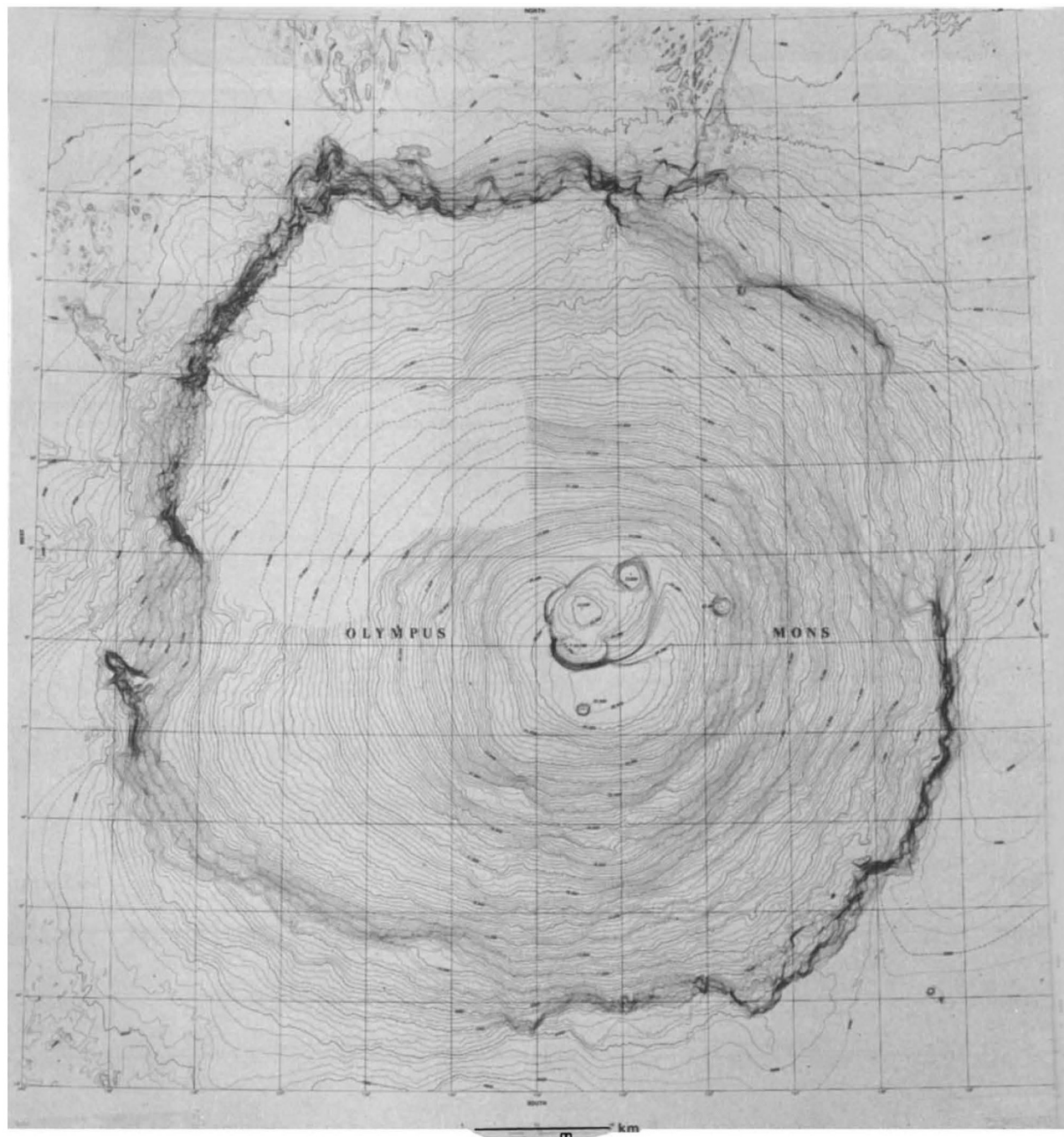


Fig. 1 Topographic map of Olympus Mons. Scale of the map, 1:1,000,000; contour interval, 200 m. Contours and elevations are in metres and are referred to the Mars topographic datum. A transverse Mercator projection was used with the central meridian placed at 134°W long.

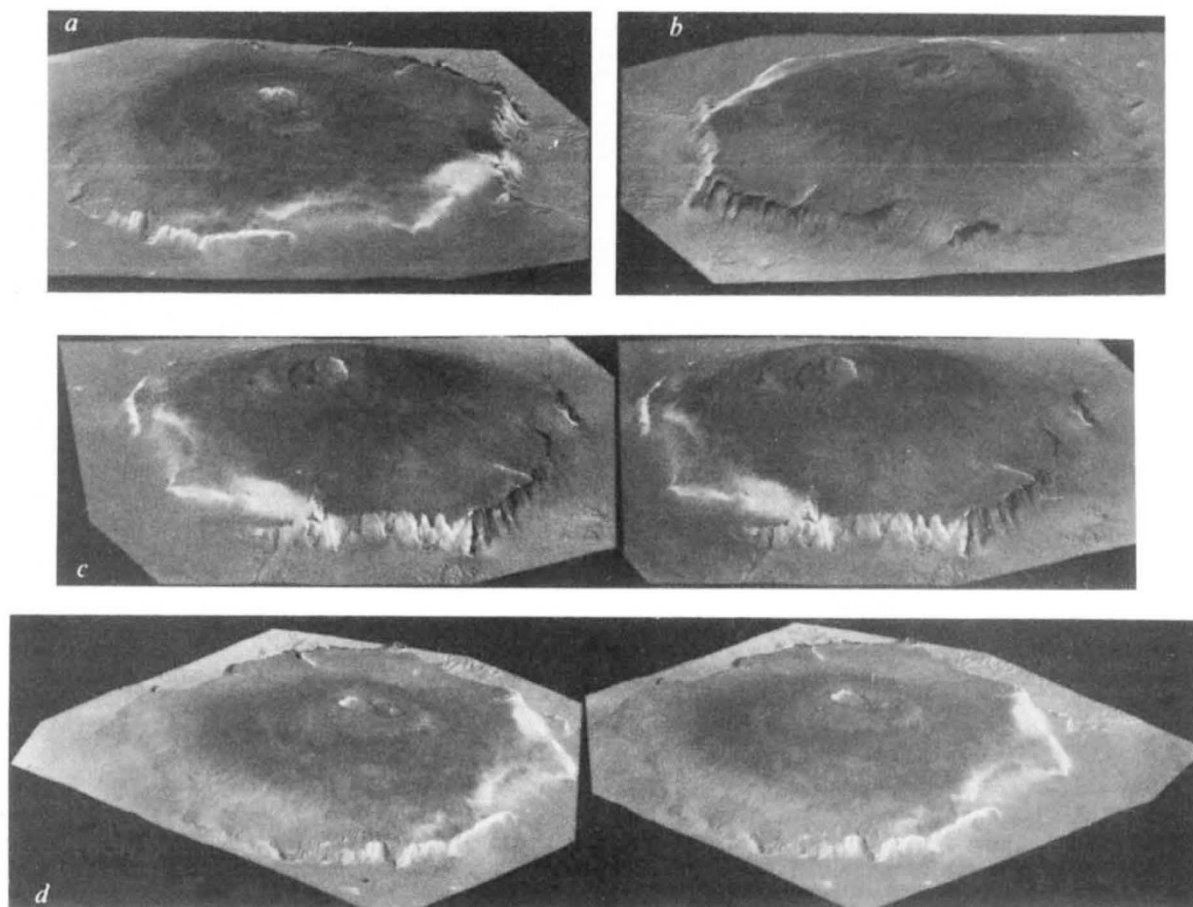


Fig. 2 Perspective views of Olympus Mons. Pictures were generated by digital image processing, from Viking Orbiter photograph 646A28. Projection is controlled by digital elevation data collected at the time of the map compilation (Fig. 1). Viewing angle is 25° from the horizon. Vertical exaggeration, $\times 5$. *a, b*, Monoscopic views from due east and due west; *c* and *d* are stereoscopic views from due north and from the south-east, respectively. Both stereo pairs have 5° separation in azimuth between the left and the right pictures. A stereoscope may be used for stereoscopic viewing.

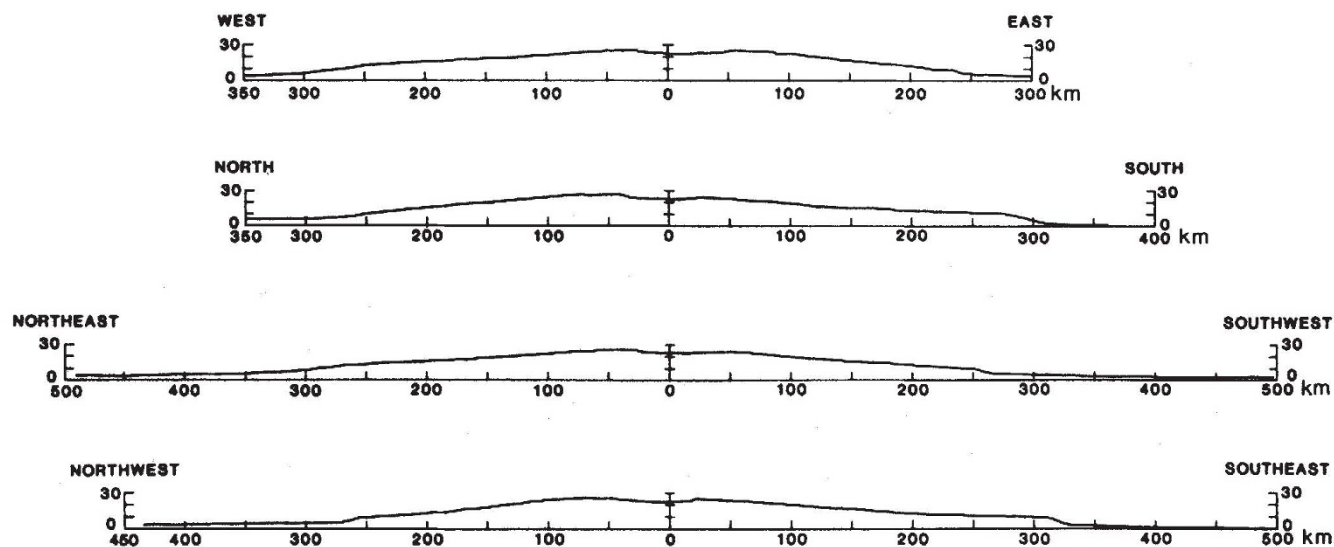


Fig. 3 Profiles measured from the topographic map (Fig. 1) of Olympus Mons. All are drawn so that they pass through the approximate centre of the summit caldera at 18.4° N lat. and 133.45° W long.

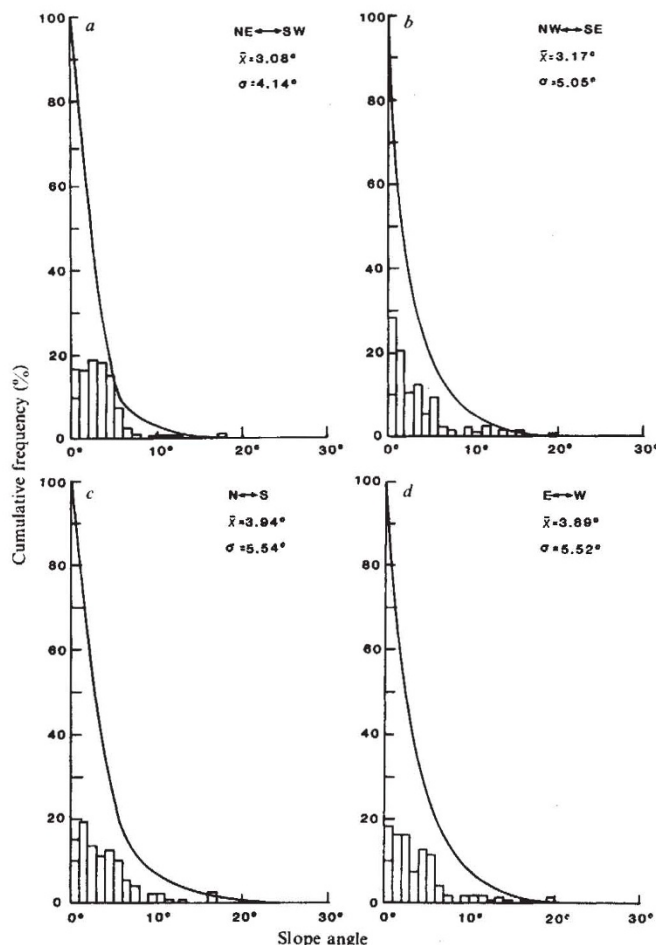


Fig. 4 Slope-frequency distributions of Olympus Mons flanks. Bars represent percentage of samples, by 1° increments of slope angle. Curved lines indicate cumulative frequency of slope angle. The quantity $\bar{x}$ is the mean absolute slope and σ is the algebraic standard deviation of the slope. *a-d*, Slopes measured along NE-SW, NW-SE, N-S and E-W directions respectively. The slope base is 5 km. These profiles, which were derived using a method of terrain analysis employing slope-frequency distributions^{23,24}, show the mean absolute slopes measured along the NE-SW, NW-SE, N-S and E-W directions, to be 3.08° , 3.17° , 3.94° and 3.89° , respectively. The algebraic standard slopes are 4.14° , 5.05° , 5.54° and 5.52° . Although the flank surface appears slightly smoother along its NE-SW direction, the surface of Olympus Mons shield is generally rough.

above the plains. Beyond the scarp, curvilinear troughs and ridges form a grooved terrain, the aureole deposits, which surround Olympus Mons. These deposits are some of the most puzzling features of the Olympus Mons region^{1,21}.

Figure 3 shows profiles measured from the map along the east-west, north-south, north-east-south-west and south-east-north-west directions. Slopes are divided, by elevation, into three groups. Within the 2–7-km interval, slopes of 2° – 3° prevail; slopes of 8° – 24° , between 7 and 16 km, are steeper mainly because there are several steep scarps within this elevation range; upper slopes, from 13 km to the outside rim of the caldera at 24 km (refs 2,22), measure 2.5° – 6.5° .

This study was performed under NASA contracts W-14585 and L-9718.

Received 29 December 1983; accepted 30 March 1984.

1. Carr, M. H., Greeley, R., Blasius, K. R., Guest, J. E. & Murray, J. B. *J. geophys. Res.* **82**, 3985–4015 (1977).
2. Wu, S. S. C., Garcia, P. A., Jordan, R. & Schafer, F. J. *3rd Int. Colloq. Mars*, LPI Contr. 441, 287–289 (1981).
3. Wu, S. S. C., Ellassal, A. A., Jordan, R. & Schafer, F. J. *Planet. Space Sci.* **30**, 45–55 (1982).
4. Batson, R. M., Hall, D. G. & Edwards, K. *NASA TM-80339*, 415 (1979).
5. Arthur, D. W. G. *Photogram. Rec.* **8**, 617–630 (1976).

6. Blasius, K. R. *J. geophys. Res.* **78**, 4411–4423 (1973).
7. Blasius, K. R., Roberts, W. J., Cutts, J. A., Duxbury, T. C. & Glackin, D. L. *10th A. Div. planet. Sci./Am. Ass. Adv. Sci. Meet.*, Pasadena, 4 (1978).
8. Blasius, K. R. & Cutts, J. A. *Icarus* **45**, 87–112 (1981).
9. Davies, M. E. *Icarus* **21**, 230–235 (1974).
10. Davies, M. E. & Arthur, D. W. G. *J. geophys. Res.* **78**, 4355–4394 (1973).
11. Hord, C. W., Barth, C. A. & Stewart, A. I. *Icarus* **17**, 443–456 (1972).
12. Wu, S. S. C., Schafer, F. J., Nakata, G. M., Jordan, R. & Blasius, K. R. *J. geophys. Res.* **78**, 4405–4410 (1973).
13. Wu, S. S. C. *Int. Rev. Ann. Geophys. Cent. Nat. Rech. Scient. AGEPA* 7-37-(1)-1-259, 147–160 (1981).
14. *Topographic Map of Mars* US Geological Survey, 1-961, (1976).
15. Wu, S. S. C. *US geol. Surv. Interagency Rep.* 63 (1975).
16. Wu, S. S. C. *Icarus* **33**, 417–442 (1978).
17. Carr, M. H. *J. geophys. Res.* **78**, 4049–4062 (1973).
18. Plescia, J. B. & Saunders, R. S. *NASA TM-80339*, 241–243 (1979).
19. Carr, M. H. *Scient. Am.* **234**, 32–43 (1975).
20. Morris, E. C. *3rd Int. Colloq. Mars*, LPI Contr. 441, 161–162 (1981).
21. Lopes, R. M. C. & Guest, J. E. *NASA TM-81776*, 176–178 (1976).
22. Wu, S. S. C., Garcia, P. A., Jordan, R. & Schafer, F. J. *NASA TM-84211*, 141–143 (1981).
23. Pike, R. J. & Rozema, W. J. *Ann. Ass. Am. Geogr.* **65**, 499–516 (1975).
24. Rowan, L. C., McCauley, J. F. & Holm, E. A. *US geol. Surv. Prof. Pap.* 599-G, 32 (1971).
25. Beatty, J. K. *Sky Telescope* 420–422 (1982).
26. Hiller, K. H., Janle, P., Neukum, G. P. O., Guest, J. E. & Lopes, R. M. C. *J. geophys. Res.* **87**, 9905–9915 (1982).

Can X-ray diffraction distinguish between protium and deuterium atoms?

Paul Seiler, Bruno Martinoni & Jack D. Dunitz

Organic Chemistry Laboratory, Swiss Federal Institute of Technology, ETH-Zentrum, CH-8092 Zürich, Switzerland

Isotopes are not normally considered to be distinguishable by X-ray crystal structure analysis. There is, of course, no problem as far as neutron diffraction is concerned, as neutron scattering amplitudes vary markedly and irregularly from one nuclide to another¹. In particular, the neutron scattering amplitudes of protium and deuterium have opposite signs so that these two isotopes are especially easily identified. Indeed, in a key experiment nearly 20 yr ago, this difference was used together with the anomalous neutron scattering of ^6Li to establish the absolute configuration of an enzymatically formed α -monodeuterioglucolic acid². We discuss here the possibility of distinguishing deuterium from protium by X-ray diffraction on the basis of the difference in the vibrational behaviour of the two isotopic species.

Least-squares analysis of sufficiently accurate and extensive diffraction data provides information not only about the atomic positions but also about the vibrational behaviour. This information is usually expressed as the components of a tensor U_i for each atom, whereby the product $I^T U_i I$ is the mean-square vibrational amplitude of the atom i about its equilibrium position in the direction of the unit vector I (ref. 3). For light X-ray scatterers, such as hydrogen atoms, the tensors U_i are usually contracted into scalars U_i (isotropic vibrational parameters) which represent the spherically-averaged motion.

Examination of the results of recent low-temperature X-ray crystal structure studies indicates that isotropic U values (mean-square vibrational amplitudes) of hydrogen atoms in small to medium-sized organic molecules are typically in the range 150 – $300 \times 10^{-4} \text{ \AA}^2$ at temperatures around 100 K (refs 4–10). They are usually larger than equivalent isotropic U values of the heavier atoms, partly because of the effect of internal molecular motions and partly because the hydrogen atoms tend to be situated on the outside of the molecules and hence further from the molecular rigid-body librational axes. The corresponding reported standard deviations of U_H (as derived by least-squares analysis) are typically ~ 30 – $40 \times 10^{-4} \text{ \AA}^2$ for reasonably good quality 100 K measurements and are perhaps as low as, say, 15 – $20 \times 10^{-4} \text{ \AA}^2$ for data sets of the highest quality available.

This is about the precision level that would be required to distinguish isotropic U of protium from that of deuterium at the 2σ (98% probability) level. For example, from tables of